Problem 1: Poisson estimation

Let $S = \sum_{i=1}^{N} x_i$, with $N > 0$. Use the shape–rate convention

$$p(\lambda) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda}, \qquad a, b > 0.$$

If b is instead the Gamma scale, replace b by $1/b$ in the results below.

(a) Joint likelihood

Independence gives

$$L(\lambda; \mathbf{x}) = \prod_{i=1}^{N} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} = \frac{\lambda^S e^{-N\lambda}}{\prod_{i=1}^{N} x_i!}.$$

(b) Maximum likelihood

$$\ell(\lambda) = S \log \lambda - N\lambda - \sum_{i=1}^{N} \log(x_i!),$$

$$\ell'(\lambda) = \frac{S}{\lambda} - N, \qquad \ell''(\lambda) = -\frac{S}{\lambda^2}.$$

For $S > 0$, setting the derivative to zero yields the unique maximum

$$\lambda_{\mathrm{ML}} = \frac{S}{N}.$$

If $S = 0$, the likelihood decreases with λ. Its maximum is at zero if the boundary is allowed; on a strictly positive parameter space, there is only a supremum as $\lambda \downarrow 0$.

(c) Maximum a posteriori

$$\log p(\lambda \mid \mathbf{x}) = C + (S + a - 1) \log \lambda - (N + b)\lambda,$$

$$\frac{d}{d\lambda} \log p(\lambda \mid \mathbf{x}) = \frac{S + a - 1}{\lambda} - (N + b).$$

For $S + a > 1$, the second derivative is negative and

$$\lambda_{\mathrm{MAP}} = \frac{S + a - 1}{N + b}.$$

For $S + a = 1$ the density decreases; for $S + a < 1$ it diverges near zero. The boundary mode is conventionally reported as zero, with the strictly positive domain qualification above.

(d) Posterior distribution

Bayes' rule gives

$$p(\lambda \mid \mathbf{x}) \propto L(\lambda; \mathbf{x}) p(\lambda) \propto \lambda^{S+a-1} e^{-(N+b)\lambda}.$$

Normalizing this Gamma kernel,

$$p(\lambda \mid \mathbf{x}) = \frac{(N + b)^{S+a}}{\Gamma(S + a)} \lambda^{S+a-1} e^{-(N+b)\lambda}, \qquad \lambda \mid \mathbf{x} \sim \mathrm{Gamma}(S + a, N + b).$$

(e) Posterior mean and variance

$$\mathbb{E}[\lambda \mid \mathbf{x}] = \frac{S+a}{N+b}, \qquad \mathrm{Var}(\lambda \mid \mathbf{x}) = \frac{S+a}{(N+b)^2}.$$

The mean can be written as

$$\mathbb{E}[\lambda \mid \mathbf{x}] = \frac{N}{N+b}\lambda_{\mathrm{ML}} + \frac{b}{N+b}\frac{a}{b}.$$

It averages the sample mean and prior mean a/b. For an interior mode, the mean exceeds λ_{MAP} by $1/(N+b)$. With fixed a, b, increasing N makes the prior influence vanish; the mean and mode approach the MLE, and the variance decreases when the sample mean remains bounded.

Problem 2: Ridge regression

Condition on X and write $y = Xw+\varepsilon$, where w is fixed, $\mathbb{E}[\varepsilon \mid X] = 0$, and $\mathrm{Cov}(\varepsilon \mid X) = \sigma^2 I_n$. Let $A = X^T X+\lambda I_d$. At $\lambda = 0$, inverse formulas require full column rank.

(a) Mean and covariance

$$w_{\mathrm{RR}} = A^{-1}X^T y = A^{-1}X^T Xw + A^{-1}X^T\varepsilon,$$
$$\mathbb{E}[w_{\mathrm{RR}} \mid X] = A^{-1}X^T Xw = A^{-1}(A-\lambda I)w = (I-\lambda A^{-1})w.$$

Thus the bias is $-\lambda A^{-1}w$. Since A is symmetric,

$$\begin{aligned}\mathrm{Cov}(w_{\mathrm{RR}} \mid X) &= A^{-1}X^T \mathrm{Cov}(\varepsilon \mid X)X(A^{-1})^T \\ &= \sigma^2 A^{-1}X^T XA^{-1}.\end{aligned}$$

This is a $d \times d$ covariance matrix. At $\lambda = 0$, the mean is w and the covariance is $\sigma^2(X^T X)^{-1}$. To express the covariance in an equivalent form, write $B = X^T X$ and

$$Z = (I+\lambda B^{-1})^{-1} = (B+\lambda I)^{-1}B = A^{-1}B.$$

Since $Z^T = BA^{-1}$,

$$\sigma^2 ZB^{-1}Z^T = \sigma^2 A^{-1}BB^{-1}BA^{-1} = \sigma^2 A^{-1}BA^{-1} = \mathrm{Cov}(w_{\mathrm{RR}} \mid X).$$

Thus the derived covariance is exactly $\sigma^2 Z(X^T X)^{-1}Z^T$.

(b) Ridge as shrinkage of least squares

For full column rank, let $X = USV^T$ be the thin SVD, with $S = \mathrm{diag}(s_1, \ldots, s_d)$ and $U^T U = V^T V = I$.

$$\begin{aligned}X^T X &= VS^2V^T, \\ X^T X+\lambda I &= V(S^2+\lambda I)V^T, \\ (X^T X+\lambda I)^{-1} &= V(S^2+\lambda I)^{-1}V^T, \\ w_{\mathrm{LS}} &= (X^T X)^{-1}X^T y = VS^{-1}U^T y, \\ V^T w_{\mathrm{LS}} &= S^{-1}U^T y, \\ U^T y &= SV^T w_{\mathrm{LS}}.\end{aligned}$$

Substituting the final equality into the ridge solution,

$$\begin{aligned}w_{\mathrm{RR}} &= V(S^2+\lambda I)^{-1}SU^T y \\ &= V(S^2+\lambda I)^{-1}S^2V^T w_{\mathrm{LS}} \\ &= \sum_{j=1}^{d} \frac{s_j^2}{s_j^2+\lambda}\, v_j(v_j^T w_{\mathrm{LS}}).\end{aligned}$$

The component along v_j is shrunk by $s_j^2/(s_j^2+\lambda)$, so smaller singular values receive stronger shrinkage. In this basis,

$$\mathrm{Cov}(w_{\mathrm{RR}} \mid X) = \sigma^2 V\,\mathrm{diag}\left(\frac{s_j^2}{(s_j^2+\lambda)^2}\right)V^T.$$

For rank-deficient X, use the compact rank-r SVD and $w_{\mathrm{LS}} = X^+ y$; the same sum over positive singular values holds for $\lambda > 0$. Null-space components are zero in both ridge and minimum-norm least squares.

Problem 3, Part 1: Linear regression

The four CSV files were extracted from the supplied ZIP. The training and test feature matrices have sizes 350×7 and 42×7. Column 7 is the constant one. The supplied linear columns are retained, and all seven coefficients, including the intercept, are penalized as specified. Part 1 is implemented in `question3.py`, using only training observations to solve

$$L(w) = \lambda \|w\|_2^2 + \sum_{i=1}^{350} (y_i - x_i^T w)^2.$$

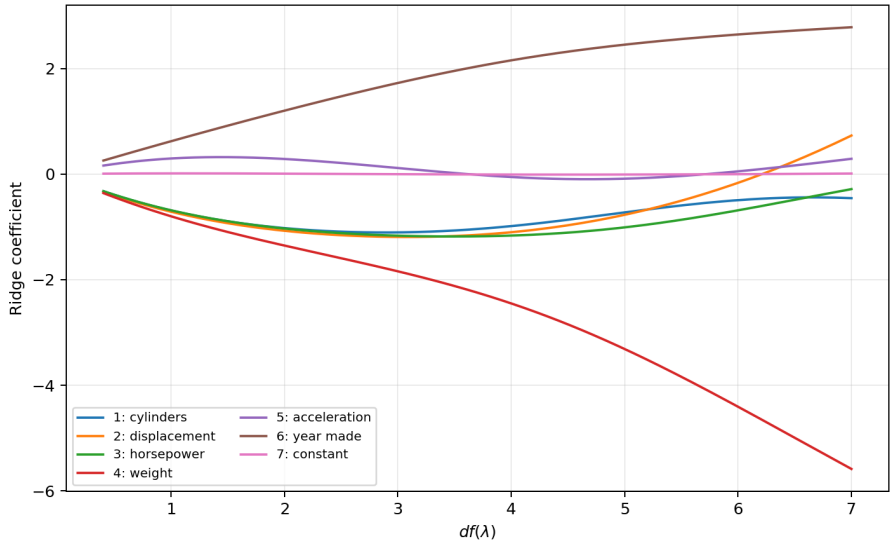
The stationarity condition gives $(X^T X + \lambda I)w = X^T y$. Test data are used only for the requested prediction error evaluation in (c).

(a) Coefficient paths

Using the training SVD,

$$w(\lambda) = V \operatorname{diag}\left(\frac{s_j}{s_j^2 + \lambda}\right) U^T y,$$
$$H(\lambda) = X(X^T X + \lambda I)^{-1} X^T,$$
$$\mathrm{df}(\lambda) = \operatorname{tr} H(\lambda) = \sum_j \frac{s_j^2}{s_j^2 + \lambda}.$$

Every integer $\lambda = 0, \ldots, 5000$ is evaluated. At zero, the SVD pseudoinverse gives least squares. Increasing λ decreases degrees of freedom; individual coefficients need not vary monotonically.



(b) Outstanding dimensions

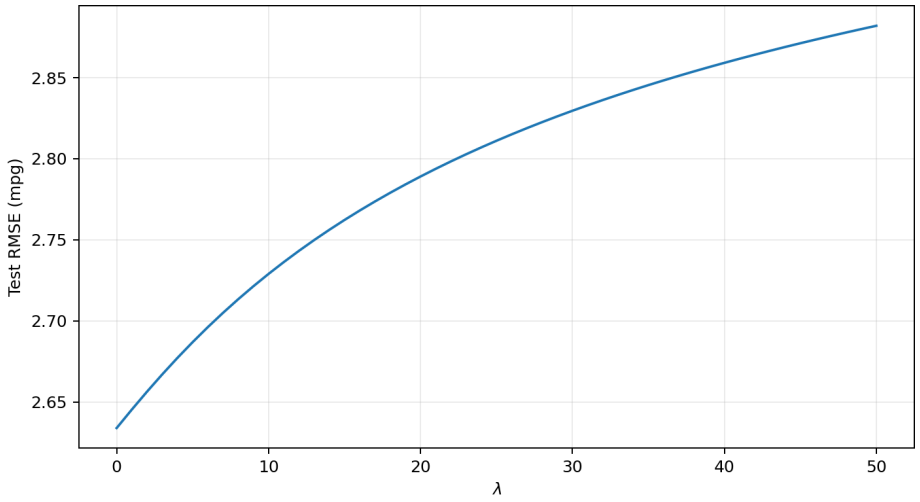
Dimensions 4 (weight) and 6 (year made) stand out. At $\lambda = 0$, their coefficients are -5.58558 and $+2.78140$; the remaining magnitudes are below 0.74. Holding the other predictors fixed, higher standardized weight predicts lower mpg and later year predicts higher mpg. These are conditional predictive associations, not causal effects. Correlated features also limit interpretation based on coefficient magnitude alone. The least squares vector, in input order, is

$$w_{\mathrm{LS}} = \begin{bmatrix} -0.456272 & 0.730192 & -0.284648 & -5.585580 & 0.289570 & 2.781398 & 0.010156 \end{bmatrix}^T.$$

(c) Linear-model test RMSE

For every $\lambda = 0, \ldots, 50$, predict all 42 test cases using

$$\widehat{y}_{\text{test}} = X_{\text{test}} w(\lambda), \qquad \text{RMSE}(\lambda) = \sqrt{\frac{1}{42} \sum_{i=1}^{42} (y_{\text{test},i} - \widehat{y}_{\text{test},i})^2}.$$



The minimum on the requested grid is at $\lambda = 0$: RMSE $= 2.633889$ mpg. Error increases with λ on this grid. Thus least squares is preferred for the linear model on these test observations; positive shrinkage does not improve its observed prediction error.

Problem 3, Part 2: Polynomial regression

(d) Construction and test comparison

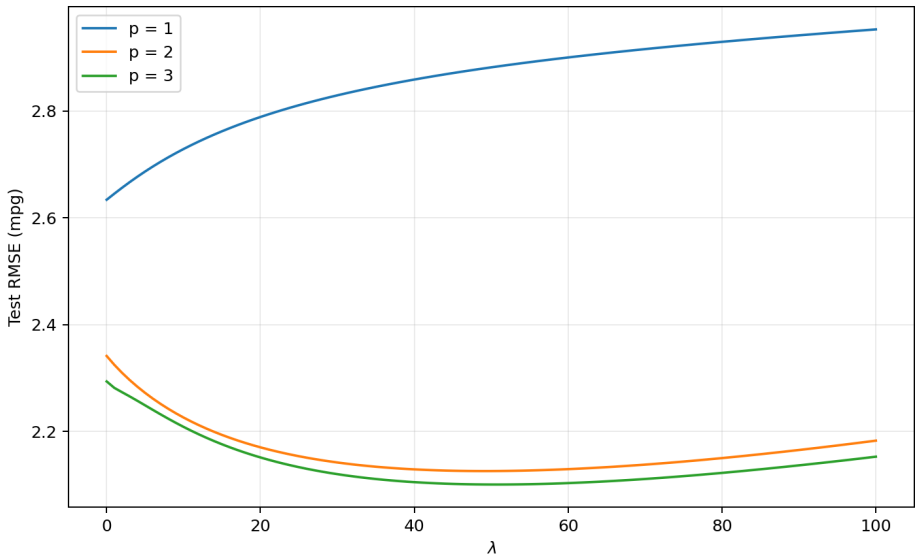
Part 2 is implemented in `question3.py`, where the Part 2 section expands the design matrix and reuses the shared SVD ridge solver. Use separate powers of each feature, without interactions:

$$\phi_p(x) = \left[x_1, \ldots, x_6, \widetilde{x_1^2}, \ldots, \widetilde{x_6^2}, \ldots, \widetilde{x_1^p}, \ldots, \widetilde{x_6^p}, 1\right]^T.$$

For $p = 1$, only the supplied linear columns and intercept are used. The expansion uses separate powers, with no interaction terms and $6p + 1$ columns. Each added column z is standardized using the empirical standard deviation formula, with statistics fitted on training data only:

$$\mu_z = \frac{1}{350}\sum_i z_i, \qquad s_z = \sqrt{\frac{1}{350}\sum_i (z_i - \mu_z)^2}, \qquad \widetilde{z} = \frac{z - \mu_z}{s_z}.$$

The same μ_z, s_z transform test columns. The intercept remains one. Fit ridge for each $p = 1, 2, 3$ and integer $\lambda = 0, \ldots, 100$.



Model choice

p	Coefficients	Best λ	Best RMSE	RMSE at $\lambda = 0$
1	7	0	2.633889	2.633889
2	13	49	2.125536	2.341201
3	19	51	2.100297	2.293523

Choose $p = 3$ based on the requested test comparison, since its minimum error is smallest, at $\lambda = 51$. Its advantage over $p = 2$ is modest: about 0.02524 mpg (1.19% of the quadratic RMSE). This plot alone does not establish that the small difference will persist on new data.

The preferred penalty changes from zero for the linear model to around 50 for the polynomial models. Added powers increase flexibility and create correlated columns; shrinkage improves test performance here. For $p = 3$, regularization reduces RMSE from 2.293523 to 2.100297 mpg. The numerical penalty depends on scaling and the sum-of-squares loss, so it is not directly transferable to other formulations.

These choices describe the supplied test comparison as requested. In a fresh predictive evaluation, choose p and λ by validation or cross-validation within the training data, then evaluate on an untouched holdout set.